

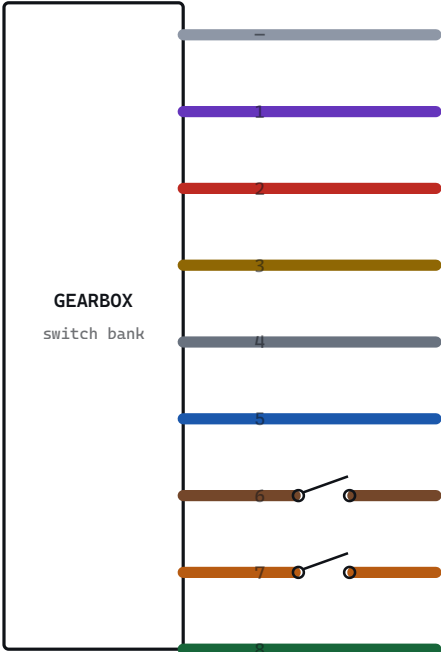
Source of truth `src/verbot/config.py`

what talks to what

```

graph TD
    USB[5V USB power bank] --> SHIM[OnOff SHIM<br/>soft power switch]
    SHIM -- "+5V" --> Pi[Raspberry Pi Zero 2 W<br/>40-pin header]
    SHIM -- "+5V motor rail - see power tree" --> DRV8833[DRV8833<br/>H-bridge, ch. A]
    Pi -- "I2C<br/>BCM 2/3" --> MCP23017[MCP23017<br/>8 buttons + LED<br/>→ front panel, 0x20]
    Pi -- "I2S<br/>BCM 18/19/21" --> MAX98357A[MAX98357A<br/>DAC + speaker<br/>→ 4-8 Ω passive speaker]
    Pi -- "PWM x2<br/>BCM 12/13 · 6 · 16" --> DRV8833
    DRV8833 -- "OUT1/OUT2" --> Gearbox[Verbot gearbox<br/>original 1984 part<br/>3V motor<br/>8 cam switches]
    Gearbox -- "8 switches<br/>9-core ribbon" --> Pi
  
```

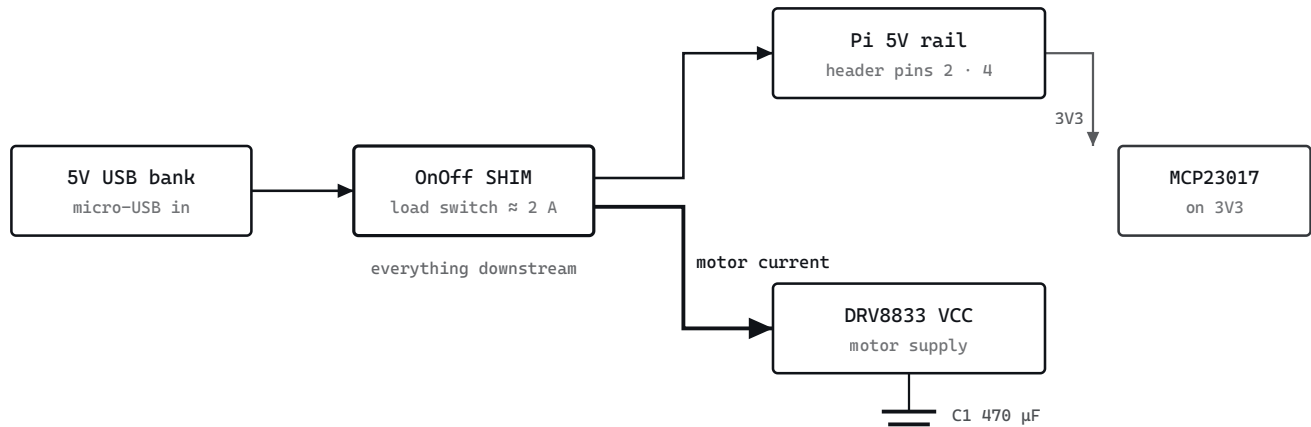
A drum inside the gearbox turns eight cams past eight normally-open switches. Each closes in a fixed sequence, telling the software which gear set is currently selected. All eight return through the single white ground.

	ORDER	WIRE	ACTION	BCM	HEADER PIN
 GEARBOX switch bank	—	WHITE	common ground return	GND	6 · 9 · 14 ...
	1	PURPLE	Stop	22	15
	2	RED	Rotate right	26	37
	3	YELLOW	Rotate left	10	19
	4	GREY	Forwards	9	21
	5	BLUE	Reverse	25	22
	6	BROWN	Put down · limit sw.	11	23
	7	ORANGE	Pick up · limit sw.	8	24
8	GREEN	Talk	7	26	

The two switches drawn with a break are the important ones. Brown and orange pass through mechanical limit switches inside the gearbox. When an arm reaches the end of its travel that circuit *opens* — the controller sees the switch release mid-action and stops the motor. Without that signal the mechanism strains against its own stop.

How an action actually runs. Drive the motor anti-clockwise until the switch for the action you want closes, then immediately reverse it. That engages a clutch: the drum stops, and a shaft inside it starts turning the selected gear set. Direction alone chooses between *hunting* and *doing*.

A single 5V bank feeds everything through the OnOff SHIM's load switch. That includes the motor, which is the part worth thinking about.



The thick line is the compromise. Motor current runs through the SHIM's load switch, rated around 2 A. A 3 V motor stalled on a 5 V rail can transiently exceed that, and the failure looks like a software hang — the Pi simply drops mid-action.

TWO THINGS THAT WILL BITE YOU

Cap the action speed. The motor is a 3 V part sitting on a 5 V rail. Keep `VERBOT_ACTION_SPEED` near ± 60 rather than ± 100 , or you are overdriving it every time it moves.

Never feed `vcc` from the Pi's 3V3 pin to get a nominal 3 V. That rail comes off the PMIC and cannot supply a stalled motor. `vcc` is the motor supply, not a logic rail — whatever you apply is what the motor sees.

DRV8833 CARRIER

CARRIER	GOES TO	BCM
VCC	+5V motor rail	—
GND	Common ground	—
IN1	pin 32	12 · PWM0
IN2	pin 33	13 · PWM1
EEP	pin 31	6
ULT	pin 36	16
OUT1 / OUT2	Verbot motor	—
IN3 · IN4 · OUT3 · OUT4	unused, channel B	—

MCP23017 · ADDRESS 0x20

BOARD	GOES TO	BCM
VDD	pin 1 · +3V3	—
VSS	Common ground	—
SDA	pin 3	2
SCL	pin 5	3
RESET	+3V3, held high	—
A0 · A1 · A2	GND → 0x20	—
GPA0–GPA7	8 panel buttons → GND	—
GPB0	330 Ω → status LED → GND	—
INTA · INTB	unused, polled at 50 Hz	—

MAX98357A

BOARD	GOES TO	BCM
VIN	+5V	—
GND	Common ground	—
BCLK	pin 12	18
LRC	pin 35	19
DIN	pin 40	21
+ / –	4–8 Ω passive speaker	—
SD	leave floating	—
GAIN	floating = 9 dB	—

ONOFF SHIM

BOARD	GOES TO	BCM
USB 5V in	Power bank	—
5V out	pins 2 · 4	—
BTN	pin 11	17
LED	pin 13	27
POWEROFF	pin 7	4

GROUND & RAILS

RAIL	HEADER PINS
+5V	2 · 4
+3V3	1 · 17
GND	6 · 9 · 14 · 20 · 25 · 30 · 34 · 39

The eight original red buttons map one-to-one onto the eight gearbox actions, exactly as the toy did in 1984. They hang off the expander's port A as active-low inputs with pull-ups enabled.

Left-to-right panel order, as assumed by <code>BUTTON_ORDER</code>			
EXPANDER PIN	BUTTON	EXPANDER PIN	BUTTON
GPA0	Stop	GPA4	Rotate right
GPA1	Forwards	GPA5	Pick up
GPA2	Reverse	GPA6	Put down
GPA3	Rotate left	GPA7	Talk

VERIFY BEFORE YOU WIRE THIS

This assumes the eight buttons are **independent switches to a common rail**. That has never been confirmed on the real panel — they may well be a matrix. Check with a meter first.

The original PCB also carried the power switching for both battery rails, so the cleanest approach is to bypass that board entirely and solder to the switch contacts directly.

The button order above is a **guess**. If a button triggers the wrong action at bring-up, reorder `BUTTON_ORDER` rather than rewiring.

Pin assignments come from `src/verbot/config.py` and `src/verbot/hardware/mcp23017.py`; the mechanical analysis and ribbon colour map were reverse-engineered by Neil Davis for *neildavis/verbot-pi*, partly from Tomy's patent US4717364A. The formal schematic lives at `hardware/verbot-schematic.pdf` — this page is its readable companion, not a replacement.